YU FU

HUMAN-AI INTERACTION | ADAPTIVE AI AGENTS | HUMAN-CENTERED AI

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RESEARCH PROFILE

I build interactive AI systems that translate complex human intent into inspectable actions while preserving meaningful human control. My work combines LLM/VLM agents, knowledge-grounded reasoning, multimodal interaction, and mixed-methods user research across presentation authoring, mental-health counseling, education, and long-term agent memory.

EDUCATION

Sichuan University-Pittsburgh Institute B.Eng. in Computer Science and Technology *Sep 2023 - Jun 2027 (expected)*
GPA: **3.84/4.00** | National Scholarship (2024) | First-Class University Scholarship (2025) | Dean's List

Tsinghua University Incoming M.S. in Data Science and Information Technology *Fall 2027*

SELECTED RESEARCH PROJECTS

ECHO: Explainable Co-editing for Presentation Slide Refinement

First Author | ACM ICMI 2026 | Published

A multimodal slide co-editing system that converts natural-language requests into inspectable, user-confirmable edit plans.

- **Research problem:** Investigated how generative editing agents can support ambiguous, object-level slide refinement without hiding consequential actions or removing user agency.
- **System design:** Designed a Plan-Confirm-Execute workflow that grounds requests in selected slide objects, exposes schema-constrained plans before execution, and supports confirmation, revision, and rollback.
- **AI architecture:** Implemented stable object grounding, adaptive LLM/VLM routing, style memory, and tool-level execution to maintain visual consistency while reducing unnecessary vision-model calls.
- **Evaluation:** Built the CoEdit-Eval benchmark and ran a controlled study with 14 participants. Compared with the baseline, task-completion time decreased by **25.8%** and NASA-TLX workload by **20.8%**.

AICo and DefenseMitigator: Human-Centered AI Counseling Systems

Third Author on DefenseMitigator | UIST 2026 Poster | AICo submitted to CHI 2027

Two systems exploring adaptive guidance and expert supervision for safer, autonomy-preserving LLM-based counseling.

- **DefenseMitigator:** Contributed to an interactive counseling system that combines a psychological-defense knowledge graph with a BDI model to identify likely defense mechanisms and select evidence-grounded guidance strategies.
- **Formative research:** Conducted and synthesized research with **6 counselors and 4 clients**; translated findings into requirements for explanation, disclosure pacing, reversibility, and preservation of client autonomy.
- **AICo:** Extended the work from one-to-one AI-user interaction to expert-supervised, multi-client collaboration. Structured expert knowledge, client context, and decision pathways so psychologists can review, revise, or override AI responses and reuse approved interventions.
- **Field deployment:** Conducted a three-week deployment with **6 licensed psychological experts**, each overseeing **5 parallel client conversations**. Analyzed behavioral logs and semi-structured interviews to study intervention timing, workload, and alignment with professional reasoning.

MathTrace: Process-Aware Analysis of Mathematical Reasoning

Core Researcher | AI for Education and HCI

A learning-analytics project that uses smart-pen traces to study how students reason, not only whether their final answers are correct.

- **Study design:** Co-designed pilot tasks and data-collection procedures for capturing timestamped handwritten problem-solving traces in realistic learning settings.
- **Process analysis:** Characterized pauses, revisions, backtracking, and transitions between scratch work and formal solutions to identify moments of uncertainty and strategy change.
- **Interface concept:** Developed a replay and diagnostic dashboard linking handwriting events to step-level evidence, feedback, and learner-model updates for process-aware AI tutors.

EduRiskX: Interpretable Early Academic-Risk Prediction

First Author | arXiv 2026

A neuro-symbolic framework that combines longitudinal behavior modeling with rule-supported explanations for early intervention.

- **Model architecture:** Integrated Transformer-based time-series modeling with F-Logic reasoning to connect predictive signals with explicit, theory-aligned explanations.
- **Early warning:** Evaluated on OULAD and identified at-risk students up to **19 weeks in advance**, earlier than strong deep baselines such as PatchTST and iTransformer, which typically reached 15-16 weeks.
- **Actionability:** Preserved strong AUC, F1, precision, and recall while generating interpretable risk factors and intervention suggestions that educators can inspect and act on.

SELECTED RESEARCH PROJECTS - CONTINUED

CassMem: Knowledge-Graph Memory for Long-Term AI Agents *Core Researcher | Submitted to WWW 2027*

A Cassandra-backed memory system for durable, inspectable, and efficient retrieval across long interaction histories.

- **Memory architecture:** Developed graph memory combining structured entities/relations with raw episodic records, supporting both interpretable traversal and high-recall semantic access.
- **Hybrid retrieval:** Integrated BM25 and dense retrieval with fusion strategies, metadata constraints, and version-aware readiness checks for safe index updates and reproducible evaluation.
- **Systems profiling:** Examined storage, retrieval latency, index consistency, and failure modes in a Cassandra-backed implementation designed for long-running adaptive agents.
- **Evaluation:** On **1,146 held-out LoCoMo queries**, Z-score fusion achieved **MRR@10 = 0.550**, compared with 0.523 for compact RRF fusion.

AgentMV: State-Guided Multi-Agent Music Video Generation *Fifth Author | AI4Art at ECCV 2026 | Accepted*

A long-horizon generation framework coordinating specialist agents through a Structured Persistent State (SPS).

- **Workflow design:** Contributed to a multi-agent pipeline spanning music understanding, narrative and script planning, visual-asset creation, segment generation, quality control, and final assembly.
- **Persistent coordination:** Used SPS to externalize shared context, intermediate decisions, resource budgets, and progress so agents can recover from errors and coordinate across long generation chains.
- **Research outcome:** The work was accepted at AI4Art at ECCV 2026 and is forthcoming in the ECCV proceedings.

Nutrient-Aware Meal Recommendation and ChefMind *First Author | Foods 2026 and arXiv 2025*

Knowledge-grounded recommendation systems for ambiguous meal requests under nutrition and safety constraints.

- Integrated USDA nutrient data with Recipe1M+ and structured food knowledge; used a 1.5B-parameter language model to translate fuzzy user intent into retrievable preferences and explicit constraints.
- Implemented an agentic critic for nutrition and safety verification, achieving **98.5% constraint satisfaction**; obtained **0.825 semantic Recall@5**, exceeding dense retrieval by 50% on complex intents.
- Developed ChefMind's Chain-of-Exploration to iteratively clarify underspecified needs, explore alternatives, and turn a vague request into an actionable recipe recommendation.

RESEARCH AND INDUSTRY EXPERIENCE

Tsinghua University, Pervasive Interaction Lab Research Intern *Mar 2026 - Aug 2026*

- Developed a digital-employee agent for graduate-admissions consulting, integrating automated WeChat-group replies, Feishu conversation records, and human-expert workflows. Supported latency monitoring, conversation-quality review, delayed-response behavior, and expert handoff.

Ernst and Young (EY) AI and Data Intern *Jan 2026 - Mar 2026*

- Contributed to invoice recognition and a Text-to-Slides module across product research, open-source solution evaluation, implementation, front-end rendering, testing, and integration into EYC.ai.

SELECTED PUBLICATIONS

Fu, Y., Kang, Y., Zhou, Y., Zhao, Y. *ECHO: Explainable Co-editing with Human-in-the-loop Operations for Presentation Slide Refinement*. ACM ICMI 2026, pp. 99-107. doi:10.1145/3776574.3831120.

Wu, Z., Zhang, R., Fu, Y., Kang, Y., Zhao, Y. *Automated Benchmark Construction for LLM Evaluation: A Source-Generator-Validator-Lifecycle Framework*. PRICAI 2026, accepted.

Nie, Z., Li, H., Fu, Y., Zhang, S., Lu, J., Yu, Z., Cai, J. *From Defense to Disclosure: Designing Interactive Systems with Multimodal Adaptive Guidance for LLM-Based Mental Health Counseling*. ACM UIST 2026 Poster, accepted.

Fu, Y., Cai, L., Wu, R., Kang, Y., Zhao, Y. *Nutrient-Aware Personalized Meal Recommendation Using Structured Food Knowledge and Constraint Verification*. Foods, 15(10), 1647, 2026. doi:10.3390/foods15101647.

Wang, H., Ouyang, L., Xia, C., Kang, Y., Fu, Y., Ouyang, Y. *AgentMV: A State-Guided Multi-Agent Framework for Budget-Aware Music Video Generation*. AI4Art at ECCV 2026, accepted.

Cai, L., Yu, C., Kang, Y., Fu, Y., Zhang, H., Zhao, Y. *Practices, Opportunities and Challenges in the Fusion of Knowledge Graphs and Large Language Models*. Frontiers in Computer Science, 7, 1590632, 2025.

TECHNICAL SKILLS

Programming	Python, C/C++, Java, SQL
AI and Systems	LLM/VLM APIs, RAG, knowledge graphs, multi-agent workflows, hybrid retrieval, PyTorch, FastAPI, Cassandra, Neo4j
HCI and Evaluation	Interactive prototyping, formative studies, semi-structured interviews, user-study design, behavioral-log analysis, qualitative/quantitative analysis
Languages	Chinese (native), English (IELTS 7.0)